Modeling Binary Outcomes

Read-ahead: what to expect and how to get set up

Barna Bose and Rachel Marriott

Time needed: About 5–10 minutes to read and run the setup check. If you have never installed R or RStudio, allow 20–30 minutes, or use Posit Cloud (no installation).

Your alarm rings at 8:00. Class starts at 8:30. Are you going?

You slept for four hours, it is raining, and your friend texts: "I'm skipping." Then you remember there might be a quiz. Do you get out of bed?

Your decision has two outcomes: attend (1) or skip (0). Sleep, weather, and the possibility of a quiz each change the *probability* that you show up. Logistic regression is a tool for exactly this kind of question: how do different factors relate to the probability of a yes/no outcome?

The workshop question: would you vote if your neighbors could find out?

In the August 2006 Michigan primary, Gerber, Green, and Larimer (2008) mailed some households a letter listing who on their street had voted in past elections, and promising to send an updated list *after* this election. The letter did not reveal *whom* anyone voted for, only *whether* they showed up.

Before the workshop: Would this letter have no effect, a small effect, or a large effect on turnout? What size change (in percentage points) would you consider politically meaningful? Write your guess down; we will check it together.

In the workshop we will use their data to model a yes/no outcome and translate the results into probabilities anyone can understand. Then you will apply the same steps to a second question: does national income predict the outbreak of civil war (Fearon and Laitin 2003)?

What you will learn

By the end of the workshop you will be able to:

- recognize and code a binary (0/1) outcome;
- explain why we often prefer logistic regression to a straight line for such outcomes;
- fit a logistic regression in R with `glm()`;
- calculate and plot predicted probabilities, with uncertainty; and
- interpret a first difference as a change in probability, and distinguish association from causation.

What you need to know already: How to run an R script and the basics of linear regression. No prior experience with logistic regression is needed.

Get set up (do this before the workshop)

1. Get the workshop folder. Go to the workshop page below, click the green **Code** button, choose **Download ZIP**, and unzip it.

https://github.com/rachelmarriott/pos5747-binary-workshop

The folder contains:

File	Purpose
`setup.R`	Installs required packages and tests your setup
`workshop.R`	Code for the live demonstration and hands-on exercise
`data/turnout.csv`	Voter-turnout data (Example 1)
`data/civil_war.csv`	Civil-war data (Example 2)
`README.md`	Data sources and variable definitions

2. Open it as an R project. Choose *one* option:

- **Local RStudio:** Install R (4.3 or newer) and RStudio Desktop if needed. Then double-click `workshop.Rproj` in the unzipped folder.
- **Posit Cloud (no installation):** Log in at posit.cloud, create a new RStudio project, and upload the workshop files. Keep the `data/` subfolder inside your working project directory.

3. Run the setup check. Open `setup.R` and click **Source**. The script:

- installs `ggplot2`, `marginaleffects`, and `sandwich` if missing;
- checks that the two data files are in `data/`; and
- fits a small logistic regression using a dataset built into R and draws a plot.

The heart of the test looks like this:

```
# Load packages
library(ggplot2)          # for plotting
library(marginaleffects)  # for predicted probabilities

# Fit a logit model: does car weight (wt, in 1,000 lbs) predict
# having a manual transmission (am: 1 = manual, 0 = automatic)?
# family = binomial() tells R the outcome is binary.
test_fit <- glm(am ~ wt, data = mtcars, family = binomial())

# Get predicted probabilities of a manual transmission
# for hypothetical cars weighing 1.5, 1.75, ..., 5.5
test_pred <- predictions(
  test_fit,
  newdata = datagrid(wt = seq(1.5, 5.5, by = 0.25))
)
# test_pred has one row per weight, with the prediction (estimate)
# and its 95% confidence interval (conf.low, conf.high)

# Plot the predictions
ggplot(
```

```
  test_pred,
  aes(x = wt, y = estimate,              # weight vs. predicted probability
      ymin = conf.low, ymax = conf.high) # edges of the confidence band
) +
  geom_ribbon(alpha = 0.2) +  # shaded 95% confidence interval
  geom_line()                 # S-shaped curve of predicted probabilities
```

You are ready if a downward-sloping probability curve appears in the Plots pane and the Console prints `SUCCESS: your setup works.`

If something goes wrong

Message	Fix
`Cannot find: data/turnout.csv`	Open `workshop.Rproj`, not just `setup.R`, then try again.
`there is no package called 'marginaleffects'`	Run `install.packages("marginaleffects")` in the Console, then rerun `setup.R`.
Installation asks to "compile from source"	Answer **No** if a compatible binary version is available.
Anything else	Email us the full error message before the workshop, or arrive 10 minutes early.

Please bring: A laptop with the setup working (or a Posit Cloud login), and your turnout prediction. See you there!

Sources

Fearon, James D., and David D. Laitin. 2003. "Ethnicity, Insurgency, and Civil War." *American Political Science Review* 97(1): 75–90.

Gerber, Alan S., Donald P. Green, and Christopher W. Larimer. 2008. "Social Pressure and Voter Turnout: Evidence from a Large-Scale Field Experiment." *American Political Science Review* 102(1): 33–48.